Statistical Analysis for Business Decisions

CodeVeda Internship – Task 3

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1. Introduction

Statistical analysis plays a vital role in supporting business decision-making by identifying patterns, testing assumptions, and quantifying uncertainty. This project applies various statistical techniques to the Boston Housing dataset to evaluate housing prices and understand the factors influencing property valuation.

The analysis includes hypothesis testing, probability distribution-based risk analysis, A/B testing, confidence interval estimation, and margin of error calculation.

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2. Objective

The objectives of this project are:

- Conduct hypothesis testing using T-Test and Chi-Square Test.
- Perform probability distribution analysis for risk assessment.
- Apply A/B testing to compare different groups.
- Calculate confidence intervals and margin of error.
- Generate business insights from statistical findings.

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3. Dataset Description

The project uses the Boston Housing dataset containing information about housing characteristics and property values.

Dataset Information

- Total Records: 506
- Total Features: 14

Important Variables

Variable	Description
CRIM	Crime rate by town

Variable	Description
ZN	Residential land zoning
INDUS	Non-retail business acres
CHAS	Charles River proximity (0 = No, 1 = Yes)
NOX	Nitric oxide concentration
RM	Average number of rooms
AGE	Age of properties
DIS	Distance to employment centers
TAX	Property tax rate
PTRATIO	Pupil-teacher ratio
LSTAT	Lower status population percentage
MEDV	Median house price

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4. Methodology

The following statistical techniques were applied:

1. T-Test
2. Chi-Square Test
3. Probability Distribution Analysis
4. A/B Testing
5. Confidence Interval Estimation
6. Margin of Error Calculation

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5. T-Test Analysis

Business Question

Do houses located near the Charles River have significantly different prices compared to houses not located near the river?

Hypotheses

Null Hypothesis (H_0):

There is no significant difference in house prices between river-side and non-river-side houses.

Alternative Hypothesis (H_1):

There is a significant difference in house prices between river-side and non-river-side houses.

Results

Metric	Value
T-Statistic	-3.9964
P-Value	0.0000739

Decision

Since the p-value is less than 0.05, the null hypothesis is rejected.

Interpretation

There is a statistically significant difference in house prices between houses located near the Charles River and those located elsewhere.

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6. Chi-Square Test Analysis

Business Question

Is river proximity associated with high house prices?

Hypotheses

Null Hypothesis (H_0):

River location and house price category are independent.

Alternative Hypothesis (H_1):

River location and house price category are associated.

Results

Metric	Value
Chi-Square Statistic	8.2719
P-Value	0.0040

Decision

The null hypothesis is rejected because the p-value is less than 0.05.

Interpretation

River location and high house prices are significantly associated.

7. Probability Distribution for Risk Analysis

Objective

Estimate the probability of a house having a price greater than 30.

Results

Metric	Value
Mean House Price	22.53
Standard Deviation	9.20
Probability (Price > 30)	20.84%

Interpretation

There is approximately a 20.84% probability that a randomly selected house has a price greater than 30. This indicates that premium-priced properties represent a smaller portion of the housing market.

8. A/B Testing

Groups

Group	Description
Group A	Houses not near the river
Group B	Houses near the river

Evaluation Metric

Median House Value (MEDV)

Results

Metric	Value
T-Statistic	-3.9964
P-Value	0.0000739

Decision

The null hypothesis is rejected.

Interpretation

River-side houses have significantly different prices compared to non-river-side houses.

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9. Confidence Interval

Objective

Estimate the true average house price with 95% confidence.

Results

Metric	Value
Lower Bound	21.73
Upper Bound	23.33

Interpretation

We are 95% confident that the true average house price lies between 21.73 and 23.33.

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10. Margin of Error

Result

Metric	Value
Margin of Error	±0.80

Interpretation

The estimated average house price may vary by approximately 0.80 units from the true population mean.

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11. Visualization

The following visualization was created:

- Histogram showing the distribution of house prices.

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12. Business Recommendations

1. River proximity significantly influences housing prices and should be considered during property valuation.
2. Premium-priced houses constitute approximately 21% of the housing market.
3. Real-estate investors can prioritize river-adjacent properties due to their higher valuation potential.
4. Statistical analysis confirms that location is a critical factor affecting property prices.
5. The narrow confidence interval and low margin of error indicate reliable estimates for decision-making.

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13. Conclusion

This project successfully applied statistical methods to support business decision-making using the Boston Housing dataset. Through hypothesis testing, risk analysis, A/B testing, and confidence interval estimation, meaningful insights were generated regarding the factors influencing housing prices. The findings demonstrate that river proximity has a significant impact on property values and can serve as an important consideration for investors, developers, and policymakers.